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Round 1 : Screening

Asked quants, logical reasoning and picture puzzle questions in 40 min test at E-vision. I submitted the test in 20 minutes.

Round 2 : DSA round

- There were 4 sets (A,B,C,D) of paper each having 1 easy, 1 medium and 1 hard level question. I had set C. Though I found those questions in my set to be easy to moderate level only.
- They wanted to test approach and problem solving ability. I had to type it in Notepad without use of IDE. There was flexibility for document format though they mentioned few points to be present which I covered. So my format for the document was :

1. Problem statement
2. Understanding problem statement and summarizing it in couple of line
3. Few examples to support
4. Few more examples for edge cases
5. Writing the step by step algorithm
6. Now writing the whole code in Java (Though they expected pseudo code)
7. Expected code output
8. Diagrammatic explanation (Which I couldn't do due to lack of time and I mentioned it too)

I followed this approach and the technical lead who took my interview complemented me that presentation and approach for edge case was good. Most probably it helped my interview to be scheduled at start only.

Set C questions :

Easy

1. For encoding an even-length binary string into a sequence of A, T, C, and G, we iterate from left to right and replace the characters as follows 00 is replaced with A 01 is replaced with T 10 is replaced with C 11 is replaced with G Given a binary string S of length N (N is even), find the encoded sequence.

Input Format : First line will contain T, number of test cases. Then the test cases follow. Each test case contains two lines of input. First line contains a single integer N, the length of the sequence. Second line contains binary string S of length N.

Output Format : For each test case, output in a single line the encoded sequence.

Medium 2. Chef has 33 bags that she wants to take on a flight. They weigh A, B, and C kgs respectively. She wants to check-in exactly two of these bags and carry the remaining one bag with her. The airline restrictions says that the total sum of the weights of the bags that are checked-in cannot exceed D kgs and the weight of the bag which is carried cannot exceed E kgs. Find if Chef can take all the three bags on the flight.

Input Format : The first line of the input contains a single integer T denoting the number of test cases. The description of T test cases follows. Each test case contains a single line of input, five space separated

integers A,B,C,D,E.

Output Format : For each test case, output in a single line answer "YES" if Chef can take all the three bags. You may print each character of the string in uppercase or lowercase (for example, the strings "yEs", "yes", "Yes" and "YES" will all be treated as identical).

Subtasks Subtask #1 (100 points): original constraints Sample 1: Input Output 3 1 1 1 15 5 8 7 6 15 5 8 5 7 15 6 YES NO YES

Hard 3. Serialization is the process of converting a data structure or object into a sequence of bits so that it can be stored in a file or memory buffer or transmitted across a network connection link to be reconstructed later in the same or another computer environment. Design an algorithm to serialize and deserialize a binary tree. There is no restriction on how your serialization/deserialization algorithm should work. You just need to ensure that a binary tree can be serialized to a string and this string can be deserialized to the original tree structure.

Input: A binary tree. Output: Output should be a serialised string and the same binary tree, reconstructed from the serialized string.

Example 1: Input: root = [1,2,3,null, null,4,5] Output: [1,2,3, null, null, 4,5] Example 2: Input: root = [] Output: []

- Personally, I found these question to be easy and similar to questions that were frequently asked in CodeChef weekly contests earlier and so I completed typing everything in Notepad everything in 90 minutes and mailed them pdf.

Round 3 : Technical Interview

- It lasted for about 1 hour and started with my personal introduction and projects I had done at college
- I was asked about the part of project I had worked on. How I approached to design and develop it.
- Then I was asked to run my project on my laptop and the tech lead who took my interview, she tested its UI and asked me to use swagger and postman. Also asked me a few questions on docker and AWS deployment and how I did it.
- Talked out security aspect, JWT, cookies and gave me few situations how I would handle them in my project if required
- I wasn't asked about OOPs but since I mentioned I used SOLID principles while development, asked me how I did it and asked me questions on my backend code
- Had a discussion over server side caching and how I did it in my project and pointed out the vulnerabilities in my project that I needed to be addressed
- Then I was asked about database schema of my project and she asked many situational questions like what I have to add new fields, which table to modify, what should be relations and then asked about triggers that I used
- On DB, I was asked to write join queries on her laptop for given tables and theory about views, stored procedures and DML operations, alias, delete without any clause and truncate, indexes and types and twister questions around it
- Then I was asked basic theory questions on Data structures like Stack, Queue, Tree and Recursion and a question of Valid Parenthesis on paper
- Then at last, we had a discussion over Software Development Life Cycle and how they at their company follows Sprints.

- I also asked her about a few points from their JD and had a friendly chat over day to day work
- Finally I was asked how long I am willing to work since there is no bond on which I said as long as they give me good projects and enough growth opportunities since it is a startup company which started 10 years ago on which she said if I do exceptional, might get opportunity to lead after a year. She had been working since 5 years in that company and told me about her experience in short when she was fresher and had given interview.

Round 4 : HR round

- I was asked to give a 10 minutes HR round and the HR asked me basic HR questions and she took some notes out of it.

This was my 1st interview experience on 14 th March in CCPP. I had lower number of calls and this was my 5th call in CCPP due to lower score in ccee but got selected in it. I took her outlook / feedback on my project. I had answered about 70 to 80 % of the questions correctly and the interviewer was also supportive as she gave me hints as and when needed and it was a good experience.